



Time Series Analysis and Forecasting

Seminar 5b: Multivariate GARCH Models



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Seminar Outline

- ▣ **Multiple Choice Quiz** – Knowledge check (10 questions)
- ▣ **True/False Questions** – Conceptual checks
- ▣ **Practice Problems** – Calculations and interpretations
- ▣ **Python Workflow** – DCC estimation on real data
- ▣ **AI-Assisted Exercise** – Critical thinking
- ▣ **Summary** – Key takeaways

Question 1

Why are univariate GARCH models insufficient for portfolio risk management?

- (A) They cannot model volatility clustering
- (B) They do not capture time-varying covariances between assets
- (C) They do not allow maximum likelihood estimation
- (D) They cannot use the Student-t distribution

Think about what information is needed to compute portfolio VaR...

Answer Question 1

Correct Answer: (B)

- ☐ They do not capture time-varying covariances between assets

Answer Question 1

Correct Answer: (B)

- ☐ They do not capture time-varying covariances between assets

Explanation

- ☐ Portfolio VaR: $\sigma_{p,t}^2 = w'H_t w$ requires the **full matrix** H_t
- ☐ Univariate GARCH gives only individual variances $h_{ii,t}$, not covariances $h_{ij,t}$
- ☐ Correlations increase during crises — ignoring them **underestimates risk**
- ☐ Optimal hedging requires $h_{12,t}/h_{22,t}$ — impossible without a multivariate model

 TSA_ch5b_motivation

Question 2

How many unique elements does the covariance matrix H_t have for $N = 10$ assets?

- (A) 10
- (B) 45
- (C) 55
- (D) 100

Answer Question 2

Correct Answer: (C)

▣ 55

Answer Question 2

Correct Answer: (C)

▣ 55

Calculation

- ▣ H_t is symmetric: $h_{ij,t} = h_{ji,t}$
- ▣ Unique elements: $N(N+1)/2 = 10 \times 11/2 = 55$
- ▣ Of which: 10 variances ($h_{ii,t}$) and 45 covariances ($h_{ij,t}, i \neq j$)
- ▣ This is the **curse of dimensionality** — motivation for structured models

 TSA_ch5b_curse_dim

Question 3

What property does the BEKK model guarantee by construction?

- (A) Stationarity
- (B) Positive definiteness of the matrix H_t
- (C) Constant correlations
- (D) Absence of volatility spillovers

Answer Question 3

Correct Answer: (B)

- ☒ Positive definiteness of the matrix H_t

Answer Question 3

Correct Answer: (B)

- ▣ Positive definiteness of the matrix H_t

Explanation

- ▣ BEKK: $H_t = CC' + A'\epsilon_{t-1}\epsilon'_{t-1}A + B'H_{t-1}B$
- ▣ CC' is p.d. (if C has full rank)
- ▣ $A'MA$ is p.s.d. for any p.s.d. M
- ▣ Sum of p.d. and p.s.d. matrices is p.d. $\Rightarrow H_t$ is p.d. automatically!
- ▣ **Note:** stationarity requires a separate condition on the spectral radius

 TSA_ch5b_bekk_sim

Question 4

In the CCC model, the covariance $h_{ij,t}$ is given by:

- (A) $h_{ij,t} = \rho_{ij,t} \cdot \sigma_{i,t} \cdot \sigma_{j,t}$
- (B) $h_{ij,t} = \rho_{ij} \cdot \sigma_{i,t} \cdot \sigma_{j,t}$
- (C) $h_{ij,t} = \alpha_{ij} \cdot \varepsilon_{i,t-1} \varepsilon_{j,t-1}$
- (D) $h_{ij,t} = \rho_{ij} \cdot \bar{\sigma}_i \cdot \bar{\sigma}_j$

Answer Question 4

Correct Answer: (B)

☒ $h_{ij,t} = \rho_{ij} \cdot \sigma_{i,t} \cdot \sigma_{j,t}$

Answer Question 4

Correct Answer: (B)

☐ $h_{ij,t} = \rho_{ij} \cdot \sigma_{i,t} \cdot \sigma_{j,t}$

Explanation

- ☐ CCC: $H_t = D_t R D_t$ with R **constant**
- ☐ The correlation ρ_{ij} does not change over time (hence the name “constant”)
- ☐ Covariance varies only through $\sigma_{i,t}$ and $\sigma_{j,t}$ (modeled with univariate GARCH)
- ☐ (A) would be DCC (with time-varying $\rho_{ij,t}$), (D) would be the unconditional covariance

 TSA_ch5b_ccc_dcc

Question 5

How many correlation parameters does DCC(1,1) require regardless of N ?

- (A) $N(N - 1)/2$
- (B) N^2
- (C) 2
- (D) $3N$

Answer Question 5

Correct Answer: (C)

- ☒ 2 parameters (a and b)

Answer Question 5

Correct Answer: (C)

- ▣ 2 parameters (a and b)

Explanation

- ▣ DCC: $Q_t = (1 - a - b)\bar{Q} + az_{t-1}z'_{t-1} + bQ_{t-1}$
- ▣ a = reaction to shocks, b = persistence — **scalars**, not matrices!
- ▣ This is the key to DCC scalability: $N = 2$ or $N = 500$, still 2 parameters
- ▣ Plus $3N$ univariate GARCH parameters (estimated separately in Stage 1)
- ▣ Total: $3N + N(N - 1)/2 + 2$ (including the unconditional correlation $\bar{Q}$)

 TSA_ch5b_dcc_estimation

Question 6

What happens to DCC correlations during crisis periods?

- (A) They decrease, because assets move independently
- (B) They remain constant, because DCC has mean reversion
- (C) They increase, because common negative shocks raise $q_{12,t}$
- (D) They become negative, because investors flee to safe assets

Answer Question 6

Correct Answer: (C)

- ☒ They increase, because common negative shocks raise $q_{12,t}$

Answer Question 6

Correct Answer: (C)

- ▣ They increase, because common negative shocks raise $q_{12,t}$

The mechanism

- ▣ Crisis: $\varepsilon_{1,t-1} < 0, \varepsilon_{2,t-1} < 0 \Rightarrow z_{1,t-1} < 0, z_{2,t-1} < 0$
- ▣ Product: $z_{1,t-1} \cdot z_{2,t-1} > 0$ (positive and large!)
- ▣ $q_{12,t}$ increases $\Rightarrow \rho_{12,t}$ increases
- ▣ **Diversification fails exactly when it is most needed**
- ▣ The aDCC model captures this effect even more strongly through the asymmetric term g

 TSA_ch5b_crisis_corr

Question 7

Which multivariate GARCH model captures volatility spillovers?

- (A) CCC — through constant correlations
- (B) DCC — through dynamic correlations
- (C) Full BEKK — through the off-diagonal elements of A and B
- (D) Diagonal BEKK — through the diagonal elements of A

Answer Question 7

Correct Answer: (C)

- ☒ Full BEKK — through the off-diagonal elements of A and B

Answer Question 7

Correct Answer: (C)

- ▣ Full BEKK — through the off-diagonal elements of A and B

Explanation

- ▣ $a_{21} \neq 0$: shock from asset 1 affects the variance of asset 2 (shock spillover)
- ▣ $b_{21} \neq 0$: past volatility of asset 1 affects current volatility of asset 2
- ▣ CCC/DCC: each variance follows univariate GARCH — **no spillovers**
- ▣ Diagonal BEKK: A, B diagonal $\Rightarrow$ spillovers eliminated
- ▣ **Cost**: full BEKK is feasible only for $N \leq 5$

 TSA_ch5b_bekk_spillover

Question 8

The optimal minimum-variance hedge ratio is:

- (A) $h_t^* = h_{11,t}/h_{22,t}$
- (B) $h_t^* = h_{12,t}/h_{22,t}$
- (C) $h_t^* = h_{12,t}/h_{11,t}$
- (D) $h_t^* = \rho_t \cdot \sigma_{1,t}/\sigma_{2,t}$ only if correlation is constant

Answer Question 8

Correct Answer: (B)

☐ $h_t^* = h_{12,t}/h_{22,t}$

Answer Question 8

Correct Answer: (B)

☐ $h_t^* = h_{12,t}/h_{22,t}$

Proof

- ☐ Hedged portfolio: $r_{\text{hedged}} = r_{\text{spot}} - h \cdot r_{\text{futures}}$
- ☐ Variance: $\text{Var}(r_{\text{hedged}}) = h_{11,t} - 2h \cdot h_{12,t} + h^2 \cdot h_{22,t}$
- ☐ Minimize: $\frac{\partial}{\partial h} = -2h_{12,t} + 2h \cdot h_{22,t} = 0$
- ☐ $h_t^* = h_{12,t}/h_{22,t} = \rho_t \cdot \sigma_{1,t}/\sigma_{2,t}$
- ☐ With DCC: ρ_t time-varying $\Rightarrow$ more precise hedging during crises!

Question 9

Which test checks whether correlations are constant (CCC vs DCC)?

- (A) ARCH-LM test
- (B) Augmented Dickey-Fuller test
- (C) Engle–Sheppard test for constant correlation
- (D) Ljung-Box test on returns

Answer Question 9

Correct Answer: (C)

- ☒ Engle–Sheppard (2001) test for constant correlation

Answer Question 9

Correct Answer: (C)

- ☐ Engle–Sheppard (2001) test for constant correlation

Procedure

- ☐ H_0 : correlations are constant (CCC adequate)
- ☐ H_1 : correlations vary over time (DCC needed)
- ☐ Regress $\text{vech}(z_t z_t' - \hat{R})$ on lagged values
- ☐ Test statistic: $T \cdot R^2 \sim \chi^2$ under H_0
- ☐ **Empirically:** CCC is almost always rejected in financial data

 TSA_ch5b_ccc_test

Question 10

Which model is suitable for a portfolio of 100 stocks?

- (A) VEC(1,1)
- (B) Full BEKK
- (C) DCC or Factor GARCH
- (D) Diagonal BEKK

Answer Question 10

Correct Answer: (C)

☒ DCC or Factor GARCH

Answer Question 10

Correct Answer: (C)

☒ DCC or Factor GARCH

Justification

- ☐ VEC: $\sim 25,500$ parameters — completely infeasible
- ☐ Full BEKK: $\sim 20,000$ parameters — infeasible
- ☐ Diagonal BEKK: $\sim 5,250$ parameters — difficult
- ☐ DCC: $302 + 2 = 304$ parameters (estimated in two stages) — **feasible**
- ☐ Factor GARCH: K factors ($K \ll N$) — even more efficient

 TSA_ch5b_model_choice

True or False?

1. The CCC model assumes that correlations between assets change over time.
2. Two-stage DCC estimation is consistent but not fully efficient.
3. Multivariate EWMA is equivalent to Scalar BEKK with $\omega = 0$ and $\alpha + \beta = 1$.
4. The diagonal BEKK model captures volatility spillovers between assets.
5. aDCC adds only one additional parameter compared to DCC.
6. DCC-based portfolio VaR is always larger than CCC-based VaR.
7. GO-GARCH works by decomposing returns into uncorrelated factors.
8. Correlations between stocks typically decrease during financial crises.

True/False Answers

1. **FALSE** — CCC = Constant Conditional Correlation. R is fixed.
2. **TRUE** — Consistent via QML. Standard errors via sandwich estimator.
3. **TRUE** — $H_t = (1 - \lambda)\epsilon_{t-1}\epsilon'_{t-1} + \lambda H_{t-1}$ with $\lambda = 0.94$.
4. **FALSE** — Diagonal BEKK: A, B diagonal, hence **no spillovers**. Only full BEKK captures spillovers.
5. **TRUE** — The parameter g for negative asymmetry. Includes DCC ($g = 0$) as a special case.
6. **FALSE** — DCC VaR is larger **during crises** (increased correlations), but can be smaller in calm periods (lower correlations). It depends on the regime.
7. **TRUE** — Uses a mixing matrix Z and factors with individual GARCH.
8. **FALSE** — Correlations **increase** during crises (contagion). Diversification fails.

Problem 1: Portfolio VaR — CCC vs DCC

Problem statement

An equally weighted portfolio ($w_1 = w_2 = 0.5$) contains two assets with:

- Daily volatilities: $\sigma_{1,t} = 2\%$, $\sigma_{2,t} = 3\%$
- CCC correlation: $\rho = 0.4$
- DCC correlation (in crisis): $\rho_t = 0.8$

Calculate:

- (a) Portfolio volatility with CCC
- (b) Portfolio volatility with DCC (in crisis)
- (c) 99% VaR (1 day) for EUR 1,000,000, both models
- (d) How much does CCC underestimate risk during a crisis?

Solution Problem 1

Answers

- ▣ (a) $\sigma_p^2 = 0.25 \times 0.04 + 0.25 \times 0.09 + 2 \times 0.25 \times 0.4 \times 0.02 \times 0.03 = 0.0445\%$
 $\sigma_p = \sqrt{0.000445} = 2.11\%$
- ▣ (b) With $\rho_t = 0.8$: $\sigma_p^2 = 0.0325 + 0.012 = 0.000565$
 $\sigma_p = \sqrt{0.000565} = 2.38\%$
- ▣ (c) VaR 99% ($z_{0.01} = 2.326$):
 - ▶ CCC: $2.326 \times 0.0211 \times 1,000,000 = \mathbf{49,069}$ EUR
 - ▶ DCC: $2.326 \times 0.0238 \times 1,000,000 = \mathbf{55,359}$ EUR
- ▣ (d) CCC underestimates VaR by $(55,359 - 49,069)/49,069 = \mathbf{12.8\%}$ in crisis!

Problem 2: Dynamic hedging with DCC

Problem statement

A fund holds stocks and hedges them with index futures. At time t :

- ▣ $\sigma_{\text{spot},t} = 1.5\%$, $\sigma_{\text{futures},t} = 1.2\%$
- ▣ CCC: $\rho = 0.90$
- ▣ DCC (crisis): $\rho_t = 0.97$

Calculate:

- ▣ (a) CCC hedge ratio: $h^* = \rho \cdot \sigma_{\text{spot}} / \sigma_{\text{futures}}$
- ▣ (b) DCC hedge ratio (crisis)
- ▣ (c) Hedging effectiveness (HE) for each model

Solution Problem 2

Answers

- ▣ (a) CCC: $h^* = 0.90 \times 1.5/1.2 = \mathbf{1.125}$
- ▣ (b) DCC: $h_t^* = 0.97 \times 1.5/1.2 = \mathbf{1.213}$
- ▣ (c) Effectiveness: $HE = \rho^2$
 - ▶ CCC: $HE = 0.90^2 = 81.0\%$ of variance eliminated
 - ▶ DCC: $HE = 0.97^2 = 94.1\%$ of variance eliminated

CCC under-hedges by $1.213 - 1.125 = 0.088$ futures units. The residual exposure is significant!

Conclusion

DCC automatically adjusts the hedge ratio to market conditions. CCC misses the increase in correlation during crises.

Problem 3: Model selection

Problem statement

A researcher estimates models for a 3-asset portfolio:

Model	Params	Log-Lik	AIC	Dynamic ρ ?
CCC	12	-3,210	6,444	No
DCC	14	-3,185	6,398	Yes
aDCC	15	-3,180	6,390	Yes
Diagonal BEKK	12	-3,200	6,424	Yes
Full BEKK	24	-3,170	6,388	Yes

- (a) Which model has the best AIC?
- (b) Can you test CCC vs DCC with an LR test? Compute the statistic.
- (c) Which model do you recommend and why?

Solution Problem 3

Answers

- ▣ (a) Full BEKK has the lowest AIC (6,388), followed by aDCC (6,390).
- ▣ (b) Yes, CCC is nested in DCC ($a = b = 0$):
 $LR = 2 \times (-3,185 - (-3,210)) = 2 \times 25 = 50$
Under H_0 : $LR \sim \chi^2(2)$. Critical value at 1%: 9.21.
 $50 \gg 9.21 \Rightarrow$ **CCC strongly rejected!**
- ▣ (c) **aDCC**: AIC close to full BEKK but with 9 fewer parameters. For $N = 3$, full BEKK is feasible, but aDCC offers a better trade-off between flexibility and parsimony. If spillovers are important $\Rightarrow$ full BEKK.

Step 1: Download data and compute returns

```
import pandas as pd
import numpy as np
import yfinance as yf
from arch import arch_model

# Download S&P 500 and DAX
sp500 = yf.download('^GSPC', start='2010-01-01', end='2024-12-31')
dax = yf.download('^GDAXI', start='2010-01-01', end='2024-12-31')

# Percentage returns
r_sp = (sp500['Close'].pct_change() * 100).dropna()
r_dax = (dax['Close'].pct_change() * 100).dropna()

# Align data on common trading days
returns = pd.concat([r_sp, r_dax], axis=1, join='inner')
returns.columns = ['SP500', 'DAX']
print(f'Observations: {len(returns)}')
```

Step 2: Rolling correlations and DCC estimation

```
# Rolling correlations with a 60-day window
rolling_corr = returns['SP500'].rolling(60).corr(returns['DAX'])

# Stage 1: Univariate GARCH for each series
res_sp = arch_model(returns['SP500'], vol='Garch', p=1, q=1,
                    dist='t').fit(dispen='off')
res_dax = arch_model(returns['DAX'], vol='Garch', p=1, q=1,
                    dist='t').fit(dispen='off')

# Standardized residuals
z_sp = res_sp.std_resid
z_dax = res_dax.std_resid

# Stage 2: Manual DCC (scalar form)
z = np.column_stack([z_sp, z_dax])
Qbar = np.corrcoef(z_sp, z_dax) # Unconditional Q
```

Step 3: DCC dynamic correlations

```
from scipy.optimize import minimize
def dcc_loglik(params, z, Qbar):
    a, b = params
    T, N = z.shape
    Qt, ll, rho_t = Qbar.copy(), 0, np.zeros(T)
    for t in range(1, T):
        Qt = (1-a-b)*Qbar + a*np.outer(z[t-1], z[t-1]) + b*Qt
        Rt = Qt / np.sqrt(np.outer(np.diag(Qt), np.diag(Qt)))
        rho_t[t] = Rt[0, 1]
        ll += np.log(np.linalg.det(Rt)) + z[t]@np.linalg.inv(Rt)@z[t] - z[t]
        @z[t]
    return 0.5 * ll, rho_t
res = minimize(lambda p: dcc_loglik(p, z, Qbar)[0],
              x0=[0.02, 0.95], bounds=[(0,0.3),(0,0.999)],
              constraints={'type':'ineq','fun': lambda p: 0.999-p[0]-p[1]})
a_hat, b_hat = res.x
_, rho_dcc = dcc_loglik([a_hat, b_hat], z, Qbar)
```

Step 4: Portfolio VaR and visualization

```
import matplotlib.pyplot as plt
COLORS = {'blue': '#1A3A6E', 'red': '#DC3545', 'green': '#2E7D32'}
# Portfolio VaR with DCC
w = np.array([0.5, 0.5])
sigma_sp = res_sp.conditional_volatility / 100
sigma_dax = res_dax.conditional_volatility / 100
port_vol = np.sqrt(w[0]**2*sigma_sp**2 + w[1]**2*sigma_dax**2
                  + 2*w[0]*w[1]*rho_dcc*sigma_sp*sigma_dax)
VaR_99 = 2.326 * port_vol * 1_000_000
# Plot
fig, ax = plt.subplots(figsize=(14, 5))
ax.plot(returns.index, rho_dcc, color=COLORS['blue'], lw=0.8)
ax.plot(returns.index, rolling_corr, color=COLORS['red'], lw=0.8, alpha=0.5)
ax.set_title('DCC vs Rolling Correlations: S&P 500 - DAX')
ax.legend(['DCC', 'Rolling 60 days'],
          loc='upper center', bbox_to_anchor=(0.5, -0.12), ncol=2)
plt.tight_layout(); plt.show()
```

AI Exercise: critical thinking

Prompt to test in ChatGPT / Claude / Copilot

"Using yfinance, download the daily closing prices of S&P 500 (^GSPC) and FTSE 100 (^FTSE) from 2015-01-01 to 2024-12-31. Compute log returns. Estimate a DCC-GARCH(1,1) model in two stages: (1) GARCH(1,1) with Student-t for each series, (2) DCC parameters (a , b) on standardized residuals. Visualize the dynamic correlations over time. Compute equally weighted portfolio VaR at 99%. Compare with CCC VaR. I want complete Python code with plots."

Critical evaluation exercise:

1. Run the prompt and critically analyze the generated code.
2. Does it estimate DCC correctly in **two stages** or does it use simultaneous MLE?
3. Does it rescale Q_t correctly to R_t ? (diagonal $\neq 1$ before rescaling)
4. Does it check the constraint $a + b < 1$? Does it report standard errors?
5. Does VaR use portfolio volatility $\sigma_{p,t} = \sqrt{w'H_t w}$, not the sum of volatilities?

Warning: AI-generated code may produce beautiful plots but *incorrect* R_t (e.g., without rescaling).

Seminar Summary

Key concepts

- ▣ **BEKK**: automatic positive definiteness, spillovers, but $\max N \leq 5$
- ▣ **CCC**: $H_t = D_t R D_t$, constant ρ — benchmark
- ▣ **DCC**: dynamic ρ_t , only 2 additional parameters — industry standard
- ▣ **aDCC**: adds correlation asymmetry (leverage)

Key formulas

$$\square \sigma_{p,t}^2 = w' H_t w; \quad h_t^* = h_{12,t} / h_{22,t}; \quad \text{VaR}_\alpha = z_\alpha \cdot \sigma_{p,t} \cdot V$$

Tip: Start with DCC-GARCH(1,1), test CCC vs DCC with Engle–Sheppard!

Homework exercises

Exercises

1. Download BET and DAX returns. Estimate DCC-GARCH(1,1). Visualize ρ_t . When is correlation at its maximum?
2. Compute daily portfolio VaR (50/50) with DCC and CCC. Compare backtesting results.
3. For Bitcoin and Ethereum, estimate DCC. Is the correlation more volatile than between stock indices?
4. Compute the dynamic hedge ratio for BET stocks hedged with index futures. Compare CCC vs DCC.
5. Estimate a bivariate diagonal BEKK for S&P 500 and EUR/USD. Are there significant spillovers?

Replication exercise: Engle (2002)

Objective

Replicate Table 1 from Engle (2002) — DCC estimation on stock indices.

1. Download daily returns of S&P 500 and DJIA (2000–2024)
2. Estimate univariate GARCH(1,1) for each series (Student-t)
3. Compute standardized residuals $\hat{z}_t$
4. Estimate DCC parameters (a, b) by maximizing $\mathcal{L}_2$
5. Visualize the dynamic correlation ρ_t and mark crisis periods
6. **Question:** is the S&P 500–DJIA correlation more stable than S&P 500–DAX? Why?

Ref: Engle, R.F. (2002). Dynamic Conditional Correlation: A Simple Class of Multivariate GARCH Models, *JBES*, 20(3), 339–350.

 TSA_ch5b_engle2002

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Thank you!

Questions?

Seminar materials are available at: <https://danpele.github.io/Time-Series-Analysis/>



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